Week 2 Operational Insight Report

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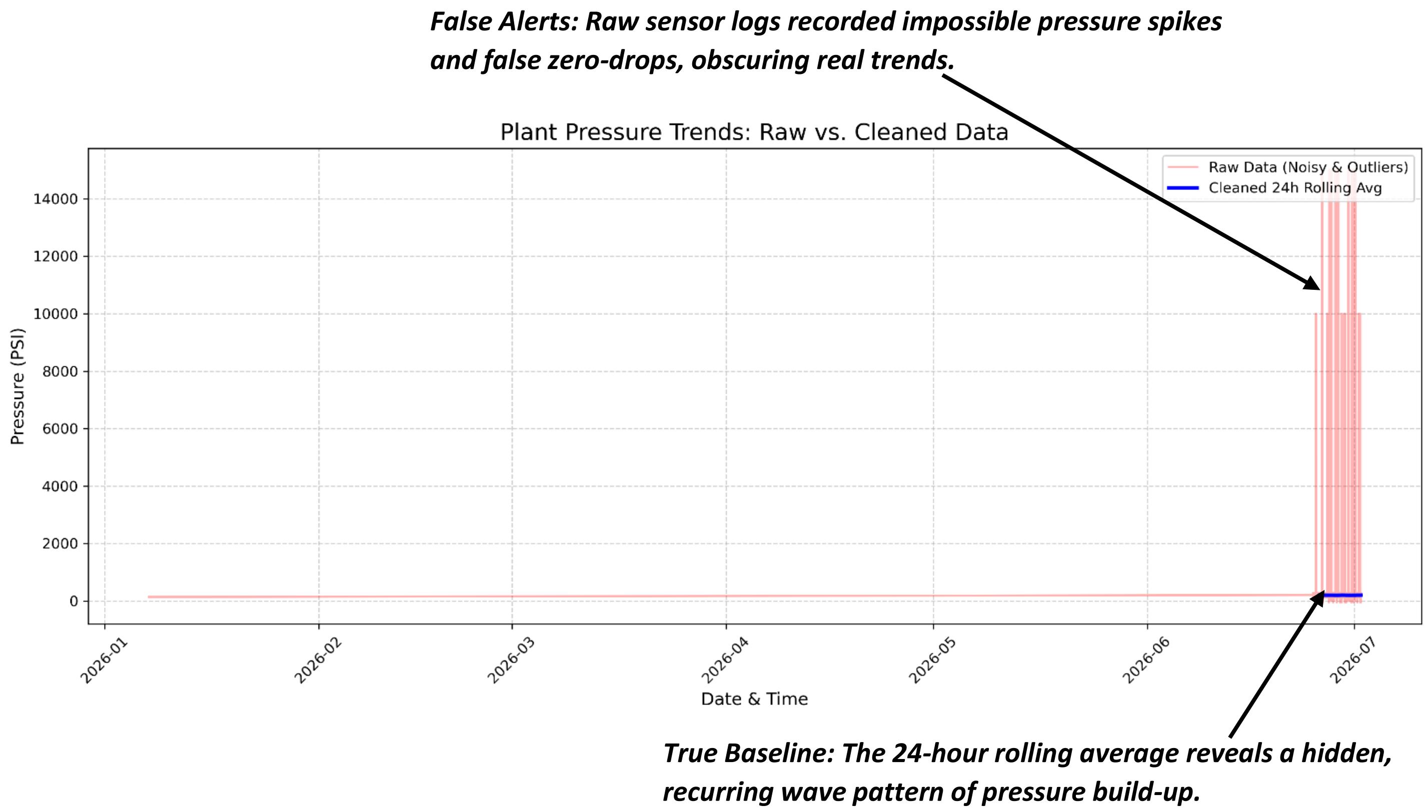
To: Plant Manager



Context

Initial evaluations of our operational sensor logs revealed severe data integrity issues that were triggering false alarms and masking true plant performance. The raw data feed contained jumbled system logs, dropped signals causing false zero-readings, and severe sensor malfunctions most notably, impossible pressure readings reaching as high as 15,000 PSI. Relying on this erratic raw data made it impossible to accurately monitor facility safety or evaluate production efficiency.

Insight

By implementing a rigorous data cleaning pipeline to filter out physical impossibilities, sort the logs chronologically, and smooth the data using a 24-hour rolling average, a critical hidden trend was exposed. Beneath the chaotic sensor noise, the true operational baseline reveals a distinct, recurring wave pattern. The system is experiencing an unprompted pressure accumulation occurring systematically every 24 hours during the night shift. This dangerous build-up was entirely invisible when looking at the raw alerts.

Action

Immediate operational intervention is required to prevent equipment fatigue or catastrophic failure. I recommend dispatching the maintenance team to inspect and recalibrate the automated pressure regulation valves in the affected zones prior to the start of tonight's shift. Furthermore, the localized sensors generating the 15,000 PSI false readings must be replaced immediately to restore trust in our real-time monitoring systems.